Activity 6 - Diagnose Hotspots and Design a Mitigation Roadmap

Alignment: K2 | K3 | K4 | A7 | LO2 Duration: 45 minutes Objective: Prioritise lifecycle hotspots and propose feasible reduction approaches without burden shifting or unsupported claims.

Scenario

Screening indicates aluminium housing, product electricity use, imported electricity, air-freighted driver boards and refrigerant leakage are leading sources. Product-use estimates are uncertain, aluminium is spend-based and a redesign could reduce mass but shorten product life. Management can fund only three initiatives this year and wants a defensible customer claim.

Materials

- Screening footprint by lifecycle stage
- Data-quality ratings and uncertainty prompts
- Reduction option facts and illustrative economics
- Hotspot and roadmap workbook

Questions

- Which sources remain priorities after considering influence and uncertainty?
- Could any option shift burdens or reduce product performance?
- Which actions are feasible within 24 months?
- What evidence supports future reduction claims?

Detailed procedure

1. Rank sources by absolute emissions and lifecycle contribution.
2. Overlay organisational influence, data confidence, trend, stakeholder importance and risk.
3. Run sensitivity checks on aluminium, product lifetime, use profile and electricity-factor assumptions.
4. Generate avoid, reduce and substitute options across design, operations, procurement, logistics, use and end of life.
5. Check every option for safety, quality, cost, feasibility, rebound, allocation and lifecycle burden shifting.
6. Estimate abatement, cost, lead time and evidence confidence without presenting forecasts as realised reductions.
7. Prioritise initiatives using transparent weighted criteria and management constraints.
8. Build 30-, 90-, 180- and 720-day milestones with owners, dependencies and decision gates.
9. Draft claim conditions linking any future statement to verified inventory evidence.

Required evidence

- Hotspot and sensitivity dashboard
- Mitigation option register
- Prioritisation scorecard

- 24-month roadmap and claim conditions

Acceptance check

Priorities reflect magnitude, influence and uncertainty; selected actions avoid undisclosed burden shifts; milestones, owners and evidence requirements are explicit; no forecast is presented as achieved reduction.

Files

- Activity-06-Workbook.xlsx - editable working file
- Activity-06-Activity-Pack.pdf - printable scenario, questions, procedure and checklist

Use fictional, anonymised or authorised workplace information only.